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Subject:**Ethical Hacking**

Subcode:**ITA1407**

## **CO2 -AT1 ASSIGNMENT**

### **Question 1: OSINT-Based Email Enumeration Assessment (20 Marks)**

Title: OSINT Email Enumeration Assessment

Objective

- Perform an OSINT assessment using theHarvester.
- Identify publicly available information related to the authorized target.
- Evaluate the security risks associated with exposed information.
- Recommend measures to reduce information leakage.

Tool Used

- Kali Linux
- theHarvester

Commands Executed

*(Include only the commands you actually ran in your lab.)*

Screenshots

- Screenshot of the command execution.
- Screenshot of the output.

## Findings

Summarize what information was discovered, such as:

- Publicly available email addresses
- Subdomains
- Hostnames
- IP addresses

Do not include sensitive personal information unless it is part of your authorized lab.

## Email Categorization

Create a table such as:

| Email       | Likely Role     |
|-------------|-----------------|
| support@... | Support         |
| sales@...   | Sales           |
| hr@...      | Human Resources |
| admin@...   | Administration  |
| tech@...    | Technical Team  |

## Risk Analysis

Explain how publicly available information can increase organizational risk, for example:

- More convincing spear-phishing emails
- Better impersonation of departments
- More effective credential-harvesting attempts
- Mapping the organization's infrastructure during reconnaissance

## Recommendations

1. Minimize publicly exposed contact information.
2. Use role-based email addresses where appropriate.
3. Provide regular security awareness training.
4. Enforce multi-factor authentication.
5. Monitor for exposed organizational information and remove unnecessary disclosures.

## Conclusion

Summarize the importance of controlling publicly available information and maintaining employee awareness.

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## **Question 2: GoPhish Awareness Simulation (20 Marks)**

Title: Phishing Awareness Simulation

### Objective

Evaluate user awareness through an authorized phishing simulation using dummy employee accounts.

### Environment

- Kali Linux
- GoPhish
- Dummy employee accounts

### Methodology

Describe:

- Creation of the email template
- Campaign setup
- Sending emails to dummy accounts
- Monitoring campaign statistics

## Screenshots

- GoPhish dashboard
- Email template
- Campaign overview
- Results dashboard

## Results

| Metric | Value |
|--------|-------|
|--------|-------|

|             |  |
|-------------|--|
| Emails Sent |  |
|-------------|--|

|           |  |
|-----------|--|
| Delivered |  |
|-----------|--|

|           |  |
|-----------|--|
| Open Rate |  |
|-----------|--|

|            |  |
|------------|--|
| Click Rate |  |
|------------|--|

Fill these values from your lab results.

## Analysis

Discuss what the results indicate about user awareness and which behaviors should be improved.

## Recommendations

- Verify sender identities.
- Avoid clicking unexpected links.
- Report suspicious emails.
- Participate in regular phishing awareness training.
- Enable multi-factor authentication.

## Conclusion

Summarize how phishing simulations improve employee awareness and reduce organizational risk.

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## **Question 3: Nmap Aggressive Scan Report (10 Marks)**

Title: Network Enumeration and Defensive Analysis

### Objective

Analyze the information obtained during an authorized network scan and discuss defensive considerations.

### Target

Authorized lab target (for example, scanme.nmap.org only if permitted by the assignment).

### Tool

- Nmap

### Command

Include the command you executed in your lab.

## Screenshots

Insert your scan output.

## Observed Information

Summarize the scan results at a high level, such as:

- Operating system information
- Open ports
- Service versions

## Security Analysis

Explain that exposed service information can help defenders prioritize patching and hardening. Discuss how organizations should verify requests made to IT support through established procedures rather than acting on unsolicited communications.

## Defensive Recommendations

1. Keep systems patched.
2. Disable unnecessary services.
3. Restrict exposed ports with firewalls.
4. Train IT staff to verify requests using official channels.
5. Monitor logs and detect unusual activity.
6. Implement network segmentation.

7. Use multi-factor authentication for administrative access.

## Conclusion

Summarize the value of regular security assessments and employee awareness in reducing organizational risk.

This structure is suitable for an assignment report while keeping the content focused on authorized cybersecurity assessment and defensive practices.